Ming-Hung Wang 王銘宏

Associate Professor
Department of Computer Science and Information Engineering, National Chung Cheng University

國立中正大學資訊工程學系 副教授

tonymhwang@gmail.com · https://tonymhwang.github.io/

Research Interests

Information Security · Computational Social Science · AI Safety

Hello, my name is Ming-Hung. I am an Associate Professor in the Department of Computer Science and Information Engineering at National Chung Cheng University (CCU). I received my Ph.D. in Electrical Engineering from National Taiwan University in 2017, after completing my B.S. in Computer Science (2008) and M.S. in Communication Engineering (2010) at National Tsing Hua University.

I lead the **Digital Society and Security Lab (DIGI-SSL)**, where our research spans **information security**, **computational social science**, and **AI safety**. We combine large language models, LLM-based agents, graph analysis, and data science to conduct empirical and theoretical studies of large-scale social media platforms and digital infrastructures.

Our work covers **information dissemination security**, **content safety**, and **platform integrity**: detecting coordinated manipulation by sockpuppets and cyber armies, identifying harmful memes and out-of-context image–text pairs, and building reliable rankings from noisy crowdsourced judgments. More recently, we have extended this work to the risks posed by generative models, such as adversarial attacks on vision-language-model-based detectors and the provenance of LLM-generated text. Through cross-disciplinary, multi-platform analysis, we aim to build a safer, fairer, and more sustainable digital society.

Our research has been published in international journals and conferences and covered by NHK, Nikkei Asia, Voice of America, and CommonWealth Magazine. Recognition includes the IEEE Tainan Section Best Young Professional Member Award and the NSTC Excellent Young Scholars Program.

經歷與服務 Experience and Service

學歷 Education

- Ph.D. in Electrical Engineering, National Taiwan University, 2017
- M.S. in Communication Engineering, National Tsing Hua University, 2010
- B.S. in Computer Science, National Tsing Hua University, 2008

經歷 Experience

國立中正大學資訊工程學系 副教授 2024/08 –
Associate Professor, Department of Computer Science and Information Engineering, National Chung Cheng University

國立中正大學巨量資料科技研究中心 主任 2025/04 –
Director, Big Data Technology Research Center, National Chung Cheng University

國立中正大學資訊處技術發展組 組長 2024/02 –
Head of Research and Development, Office of Information Technology, National Chung Cheng University

中華民國資訊安全學會 秘書長 2024/09 – 2026/01
Secretary-General, Chinese Cryptology and Information Security Association

中華民國資訊安全學會 監事 2021/09 – 2024/08
Supervisor, Chinese Cryptology and Information Security Association

中華民國資訊安全學會 教育與推廣委員會主任委員 2022/01 – 2024/08
Chairman of the Education and Promotion Committee, Chinese Cryptology and Information Security Association

國立中正大學資訊工程學系 助理教授 2021/08 – 2024/07
Assistant Professor, Department of Computer Science and Information Engineering, National Chung Cheng University

逢甲大學資訊工程學系 助理教授 2018/02 – 2021/07
Assistant Professor, Department of Information Engineering and Computer Science, Feng Chia University

逢甲大學資通安全研究中心 副主任 2020/01 – 2021/07
Deputy Director, Information and Communication Security Research Center, Feng Chia University

中華民國資訊安全學會 副秘書長 2018/05 – 2021/05
Deputy Secretary-General, Chinese Cryptology and Information Security Association

資訊工業策進會 正工程師 2017/08 – 2018/01
Senior Engineer, Institute for Information Industry

中央研究院 研究助理 2010/09 – 2013/12
Research Assistant, Academia Sinica

學術服務 Academic Service

學術會議服務與期刊編輯 Conference and Editorial Service

- Co-organizer, GenChal@INLG-2026 (Live Commentary Planning and Generation)
- Program Co-chair, AsiaJCIS 2025
- Program Chair, SICSS-Taiwan 2022
- Keynote Speaker, ARTMAN@ACM CCS 2025
- Keynote Speaker, RET 2024
- Invited Speaker, AsiaJCIS 2025
- Publication Chair, CISC 2021, 2023
- Publication Chair, ACM ASIACCS 2020
- Program Committee, AAAI 2026-2027

- Program Committee, IEEE DSC 2026
- Program Committee, ACM CIKM 2025-2026
- Program Committee, IEEE MCSoC 2025
- Program Committee, IEEE ICA 2025-2026
- Program Committee, dg.o 2025
- Program Committee, DemocrAI 2025
- Program Committee, IEEE MIPR 2023-2025
- Program Committee, AsiaJCIS 2022-2024
- Program Committee, DDRC 2021
- Program Committee, International Computer Symposium 2020, 2024
- Program Committee, The Web Conference 2020 (Poster session)
- Program Committee, CISC 2020-2024
- Guest Editor, Journal of Information Science and Engineering, 2023

期刊審稿 Journal Referee

- ACM Transactions on Knowledge Discovery from Data
- BMC Public Health
- The Computer Journal
- Cyberpsychology, Behavior, and Social Networking
- ETRI Journal
- IEEE Access
- IEEE Internet of Things Journal
- IEEE JSAC
- IEEE Sensors Journal
- IEEE Systems Journal
- IEEE Transactions on Big Data
- IEEE Transactions on Computers
- IEEE Transactions on Computational Social Systems
- IEEE Transactions on Information Forensics and Security
- IEEE Transactions on Reliability
- IEEE Transactions on Services Computing
- IEICE Transactions on Information and Systems
- Journal of Computational Social Science
- Journal of Information Security and Applications
- Journal of Information Science and Engineering
- JMIR Formative Research
- Scientific Reports
- Sensors
- Taiwan Politics

論文著述 Publications

Selected works. Full list on Google Scholar; citation counts (10+) from Google Scholar as of October 2026.

Selected Journal Articles

1. **Quantifying Perceived Toxicity in Multimodal Memes through Crowdsourced Paired Comparisons.** Ming-Hung Wang, Hsiu-Ling Chu, Jia-Ru Song, Wei-Bin Lee. *New Media & Society, 2026. (Accepted, SSCI Q1, Rank: 15/228)*
2. **AlignDC: A Scalable and Reliable Ranking Algorithm for Noisy Crowdsourced Pairwise Comparisons.** Ming-Hung Wang, Han-Ping Wu, Yu-Yu Chang, Hsiu-Ling Chu, Wei-Bin Lee. *IEEE Transactions on Reliability, 2026, 75, 2319-2333. (SCI Q1)*
3. **A Confidence-Aware Human-Machine Collaboration Framework for Harmful Meme Detection.** Xin-Yi Lin, Yu-Yao Tseng, Sih-Cin Huang, Ming-Hung Wang*. *IEEE Transactions on Computational Social Systems, 2026, 13 (3), 3825-3836. (SCI Q1)*
4. **METS: Metadata-Embedded Steganography for Digital Rights Tracking in LLM-Generated Text.** Chang-Sheng Chang, Yu-Yao Tseng, Sih-Cin Huang, Ming-Hung Wang*. *IEEE Open Journal of the Computer Society, 2026, 7, 417-429. (ESCI Q1, IF: 8.2)*
5. **Beyond Text: Detecting Image Propaganda on Online Social Networks.** Ming-Hung Wang, Yu-Lin Chen. *IEEE Transactions on Sustainable Computing, 2025, 10 (1), 120-131. (SCI Q1)*
6. **CrowdDC: Ranking from Crowdsourced Paired Comparison with Divide-and-Conquer.** Ming-Hung Wang, Chia-Yuan Zhang, Jia-Ru Song. *IEEE Transactions on Computational Social Systems, 2024, 11 (2), 3015-3021. (SCI Q1)*
7. **An SDN-Driven Reliable Transmission Architecture for Enhancing Real-Time Video Streaming Quality.** Ming-Hung Wang, Ting-Shen Hsieh, Yu-Yao Tseng, Po-Wen Chi. *IEEE MultiMedia, 2024, 31 (1), 54-64. (SCI)*
8. **Analyzing Image-based Political Propaganda in Referendum Campaigns: From Elements to Strategies.** Ming-Hung Wang, Wei-Yang Chang, Kuan-Hung Kuo, Kuo-Yu Tsai. *EPJ Data Science, 2023, 12, 29. (SCI Q1, SSCI Q1)* Cited by 24
9. **Learning to Recognize Sockpuppets in Online Political Discussions.** Nhut-Lam Nguyen, Ming-Hung Wang*, Chyi-Ren Dow. *IEEE Systems Journal, 2022, 16 (2), 1873-1884. (SCI)*
10. **Positioning and Categorizing Mass Media Using Reaction Emojis on Facebook.** Ming-Hung Wang*. *The Computer Journal, 2021, 64 (3), 451-461. (SCI)*
11. **The Evolution of Rumors on a Closed Social Networking Platform During COVID-19: Algorithm Development and Content Study.** Andrea W Wang, Jo-Yu Lan, Ming-Hung Wang, Chihhao Yu. *JMIR Medical Informatics, 2021, 9 (11), e30467. (SCI)*
12. **Blurring Political Leanings of Messages on Social Networks Using Deniable Steganography.** Ming-Hung Wang, Po-Wen Chi. *IEEE Access, 2019, 7, 87692-87703. (SCI Q1)*
13. **Estimating Ideological Scores of Facebook Pages: An Empirical Study in Taiwan.** Ming-Hung Wang, Alex Chuan-Hsien Chang, Kuan-Ta Chen, Chin-Laung Lei. *The Computer Journal, 2017, 60 (11), 1675-1686. (SCI)*
14. **SDUDP: A Reliable UDP-based Transmission Protocol over SDN.** Ming-Hung Wang, Lung-Wen Chen, Po-Wen Chi, Chin-Laung Lei. *IEEE Access, 2017, 5, 5904-5916. (SCI Q1)* Cited by 46

Selected Conference Proceedings

1. **Same Person, Different Perception: Adversarial Identity Substitution against LVLM-Based Out-of-Context Detection.** Zhi-Tang Lin, Sih-Cin Huang, Ming-Hung Wang. *ACCV 2026.*
2. **Linguistic Cues in Political Debates: Lexical Choices and Evaluative Framing in GPT-Generated Commentary.** Yu-Yu Chang, Chung-Chi Chen, Ming-Hung Wang. *AACL-IJCNLP 2026 (Findings).*
3. **DR-CUP: A Dataset on Real-time Commentary in U.S. Presidential Debates.** Yu-Yu Chang, Huan-Wen Ho, Chung-Chi Chen, Ming-Hung Wang. *LREC 2026.*
4. **MemeUMPIRE: Zero-Shot Multi-Agent Debate with Judge for Explainable Harmful-Meme Reasoning.** Sih-Cin Huang, Meng-Rong Wu, Tuang Hee Low, Tomohiro Morikawa, Harry Chandra Tanuwidjaja, Takeshi Takahashi, Ming-Hung Wang. *IEEE ICC 2026.*
5. **Sockpuppets Detection Using Markov-Driven Graph Convolutional Networks with Ensemble Learning.** Chun-I Fan, Ming-Feng Tsai, Chun-Pin Hu, Ming-Hung Wang. *IEEE DSC 2024.*
6. **DRM-SN: Detecting Reused Multimedia Content on Social Networks.** Wen-Hsiang Li, Yao-Cheng Lu, Wen-Kai Hsiao, Yu-Yao Tseng, Ming-Hung Wang. *IEEE MIPR 2024.*
7. **Advancing Stance Detection of Political Fan Pages: A Multimodal Approach.** Kuan-Hung Kuo, Ming-Hung Wang, Hung-Yu Kao, Yu-Chen Dai. *The Web Conference 2024 (WWW 2024, short paper).*
8. **Take Expert Advice Judiciously: Combining Groupwise Calibrated Model Probabilities with Expert Predictions.** Sumeet Gupta, Shweta Jain, Shashi Shekhar Jha, Pao-Ann Hsiung, Ming-Hung Wang. *ECAI 2023.* Cited by 10
9. **POSTER: How Do Suspicious Accounts Participate in Online Political Discussions? A Preliminary Study in Taiwan.** Ming-Hung Wang, Yu-Chen Dai. *ACM ASIACCS 2020 (poster).*
10. **Detecting Potential Cyber Armies of Election Campaigns Based on Behavioral Analysis.** Ming-Hung Wang, Nhut-Lam Nguyen, Chyi-Ren Dow. *Complex Networks 2018.*
11. **Infection Categorization Using Deep Autoencoder.** Ming-Hung Wang, Meng-Han Tsai, Wei-Chieh Yang, Chin-Laung Lei. *IEEE INFOCOM 2018 (poster).*
12. **Boosting Election Prediction Accuracy by Crowd Wisdom on Social Forums.** Ming-Hung Wang, Chin-Laung Lei. *IEEE CCNC 2016.* Cited by 21
13. **SDN Storage: A Stream-based Storage System over Software-defined Networks.** Ming-Hung Wang, Po-Wen Chi, Jing-Wei Guo, Chin-Laung Lei. *IEEE INFOCOM 2016 (poster).* Cited by 12
14. **Modelling Polarity of Articles and Identifying Influential Authors through Social Movements.** Ming-Hung Wang, Chin-Laung Lei. *IEEE SMC 2015.*
15. **Understanding Potential Microtask Workers for Paid Crowdsourcing.** Ming-Hung Wang, Kuan-Ta Chen, Shuo-Yang Wang, Chin-Laung Lei. *AAAI HCOMP 2013 (Works-in-Progress Track).*
16. **Privacy Crisis due to Crisis Response on the Web.** Shao-Yu Wu, Ming-Hung Wang, Kuan-Ta Chen. *IEEE TrustCom 2011.*

Other Writings

1. **便利之下的風險：不在籍投票與選民匿名性的挑戰** 王銘宏 自由開講@自由時報 2025-07-18
2. **補上人才斷鏈——嘉義市邁向南方科技樞紐的關鍵一步** 王銘宏 未來城市@天下雜誌 2025-04-29

3. **AI做事實查核有辦法比人更客觀嗎？中正資工系助理教授王銘宏：拓展範圍，但信任感仍待建立** 黃兆徽 新聞真假掰 2024-01-03
4. **【專訪】王銘宏教你三招揪出「偽」分身帳號** 蘇元和 全民查假會社 2023-12-13
5. **網軍抓到了？學者3數據，揪台灣人都怎麼罵譚德塞** 王銘宏 天下雜誌 2020-04-10
6. **螢幕後的殺手：輿論操作與對抗** 王銘宏、戴毓辰 蘋果日報 2019-12-10
7. **「抓到了？！」用數據分析鳥瞰 PTT 政治文帳號** 王銘宏 菜市場政治學 2018-10-20
8. **努力搶下百里侯寶座，三年後的期中考進步了嗎？** 王銘宏、王宏恩 菜市場政治學 2017-09-15
9. **PTT 評論行為分析 – 以太陽花學運期間為例** 王銘宏 菜市場政治學 2016-01-05

獲獎與榮譽 Honors & Awards

獎項

2025	IEEE Tainan Section Best Young Professional Member Award
	國科會優秀年輕學者研究計畫
	國立中正大學教師優良教學獎
2024	國立中正大學青年學者獎
	國立中正大學工學院青年學者獎
	國立中正大學工學院優秀論文獎
2023	臺灣綜合大學系統年輕學者創新研發成果選拔佳作獎
2022	國立嘉義大學附設實驗國民小學第二屆傑出校友
2022–2026	國立中正大學特殊優秀人才獎勵
2021	逢甲大學資電學院行政服務優良教師
2019–2021	逢甲大學學術研究獎勵
2018	科技部補助大專校院延攬特殊優秀人才
2017	中華民國資訊學會博士論文佳作獎
	國立臺灣大學電機系最佳博士論文獎

優良課程

2026	114 學年度第 1 學期國立中正大學優良課程
2025	113 學年度第 2 學期國立中正大學優良課程
2024	112 學年度第 2 學期國立中正大學優良課程
2023	111 學年度第 2 學期國立中正大學優良課程
	111 學年度第 1 學期國立中正大學優良課程

指導學生

2026	第 56 屆全國技能競賽青年組指導老師（網路安全職類佳作）
	第 56 屆全國技能競賽南區分區技能競賽青年組指導老師（網路安全職類第一名）
2025	114 年度全國大專電腦軟體設計競賽指導老師（2 佳作）
	ICPC 2025 亞洲區臺中站指導老師（1 銀牌、1 佳作）
	ICPC 2025 臺灣區線上程式競賽指導老師（3 銅牌、1 佳作）
	GiCS 2025 尋找資安女婕思指導老師（創意發想賽優勝）
2024	ICPC 2024 亞太區錦標賽指導老師（1 佳作）
	ICPC 2024 亞洲區臺中站指導老師（3 銅牌、1 佳作）
	ICPC 2024 臺灣區線上程式競賽指導老師（3 銀牌、3 銅牌）
	113 年度全國大專電腦軟體設計競賽指導老師（4 佳作）
2023	ICPC 2023 亞洲區桃園站指導老師（2 銅牌、1 佳作）
	ICPC 2023 臺灣區線上程式競賽指導老師（3 銀牌、1 佳作）
	112 年度全國大專電腦軟體設計競賽指導老師（2 佳作）
	第 33 屆全國資訊安全會議最佳簡報獎
	第 33 屆全國資訊安全會議最佳學生論文獎佳作
2021	第 31 屆全國資訊安全會議最佳學生論文獎佳作